

Study on the trap properties of nano-ZnO modified polyimide based on surface potential decay method

Bo Zhang, Jiang Wu, Xiao-Quan Zheng

State Key Laboratory of Electrical Insulation and Power Equipment

Xi'an Jiaotong University

Xi'an, China

granthuilei@163.com

Abstract—The space environment will charge the surface of the spacecraft. Due to the low discharge ability of the dielectric material, it may cause the surface discharge of the spacecraft and affect the normal work of the spacecraft. Polyimide is a typical dielectric material on the spacecraft, this paper attempts to improve the surface discharge ability by modifying polyimide with nano-ZnO. The research method is based on the surface potential decay platform, which is an intuitive approach to evaluate the states of surface charging. The surface potential decay characteristics of pure polyimide and its modified samples are tested. The surface potential decay mechanism of the dielectric is analyzed by calculating the trap characteristics. Results show that the surface potential of ZnO modified polyimide has less decrement and rate of decay than that of the pure polyimide. The deep trap density of electron trap and hole trap density is less than that of pure polyimide, the shallow trap of pure polyimide is the lowest. The change of electron trap level is small, while the hole trap level increases.

Keywords—Surface potential decay; polyimide; ZnO; electron trap; hole trap

I. INTRODUCTION

Polyimide (PI) is a kind of engineering material with excellent comprehensive performance, widely used in aerospace, aviation, and other fields. In the space environment, the charged particles interact with the surface dielectric of the spacecraft and cause the surface charging. Owing to the insulating property of dielectric, the charge on the surface of the dielectric can hardly dissipate, so that the electric field on the surface of the material is seriously distorted, further causing the decrease of flashover voltage along the surface, even induce the surface breakdown thus interfere the normal operation of spacecraft equipment and even lead to spacecraft failure [1]. Therefore, studying and improving the surface charge injection and discharge characteristics of materials is the current research hotspot.

Common methods for measuring insulation traps are the thermoelectric stimulated current (TSC), pulsed electroacoustic (PEA), and surface potential decay (SPD). However, the TSC and the PEA method cannot distinguish the electron and hole traps. In this paper, the SPD method is used to analyze the migration of carriers in polymers by applying voltage with different polarities to distinguish the electron and hole traps.

For nearly half a century, the SPD method has gradually become a powerful tool for studying the charge transport properties in polymer materials. Simmons and Tam [2,3] proposed the isothermal current decay theory by studying the trapping and detrapping of carriers in semiconductors and insulating materials in the 1970s. This method adopts charging to deposit a certain amount of charge on the surface of the material. By measuring the decay of potential induced by charge over time, the internal charge transport characteristics and trap distribution characteristics are studied. The SPD method is a convenient, low-cost method to measure the space charge and charge traps and provides relevant physical information such as charge injection, migration, relaxation processes and traps. Therefore, the SPD method is widely used to characterize trap distribution and charge transport properties in polymer materials.

Based on our previous study [4,5], the addition of ZnO modification may enhance the flashover and non-linear conductivity properties of PI, which are related to the carrier migration and trap characteristics. Therefore, the method of nano-ZnO modifying the PI composites is adopted to investigate the surface potential characteristics.

II. THEORY

In the past few decades, lots of research on SPD was reported and various models were developed to explain the potential decay characteristics.

The deposition of carriers on the dielectric is mainly influenced by three ways. That is the process of transport through materials, along with the surface and the neutralization on the gas-solid interface [6-8]. The normal direction of electric field is much higher than that of the tangential electric field and by controlling the humidity, temperature and other conditions, surface conduction and interface neutralization can be ignored. Bulk conduction in the dielectric is the main way of SPD.

The dielectric is arranged between the high voltage electrode and the grounded electrode. Due to the high local field strength at the pinpoint of the needles, corona will occur, and a large number of charged particles will be generated after air ionization. The charged particles will be accelerated by the high voltage applied on the mesh grid electrode and move towards the upper surface of the dielectric. The charge initially

accumulates on the upper surface and then transport from the surface to the inner of the dielectric. Some of the charge is trapped in the trap center and forms space charge.

The polymeric structure of the insulating material leads to discrete or continuous trap levels. In our study, the field strength was too low to result in tunneling and ionization. It is also impossible to induce photoelectron assisted detrapping only exposed to visible light, so lattice thermal vibration is the only way to generate carriers. In order to simplify the calculation, it is considered that the surface traps are uniformly distributed, and it is assumed that carriers will not re-trap due to the electric field. Based on the assumptions above, Wen-Wei Shen [9] obtains the equations as in (1) and (2). The relationship between trap level and density can be calculated.

$$E_T = kT \ln(\gamma t) \quad (1)$$

$$N(E_t) = \frac{\epsilon_0 \epsilon_r t}{k T f_0(E_t) \delta L q} \frac{dV_s}{dt} \quad (2)$$

In (1) and (2), E_T is the trap energy level, k is the Boltzmann constant, T is the absolute temperature, t is the time, $N(E_t)$ is the density of trap, ϵ_0 is the vacuum dielectric constant, ϵ_r is the relative dielectric constant, q is the electric charge, L is the specimen thickness, δ is the thickness of top charge layer, usually set as 2 μm , γ is the attempt-to escape frequency of trapped electron, set as $4.17 \times 10^{13} \text{s}^{-1}$ from theoretical modelling of infinitely deep square potential well, assume $f_0(E_t)$ as 1 for initial occupancy of electron and hole traps with negative and positive corona charging, respectively. It means that the traps are considered to be fully occupied and thus trap density equals to the density of trapped charges.

III. EXPERIMENTAL METHODS

A. Samples and preparation

The raw material of PI specimen was the SKPI-MS30 type of molding powder with the median diameter of 50 μm . The median diameter of ZnO is 50nm. ZnO surface was treated by KH550 silane coupling agent and then uniformly dispersed in the powder. The powder was molded with high temperature and high pressure. Thus, the pure PI and 1wt%, 2wt%, 3wt%, 5wt% nano-ZnO modified PI specimen was obtained. The SEM results are shown in Fig. 1. The ZnO is uniformly dispersed. The thickness of the PI is 0.25mm.

B. Corona charging setup and potential decay measurements

The physical system of SPD technique under consideration usually is that of an insulating sheet, with its free surface charged and its opposing face in good mechanical and electrical contact to an earthed back electrode.

A schematic experimental setup is shown in Fig. 2. The samples were charged on their free upper surface using a needle-grid corona charging setup which consists of a high-voltage needle electrode, a mesh grid electrode, and a movable grounded electrode. The needle electrode was charged to $\pm 12\text{kV}$, the mesh grid electrode was charged to $\pm 3\text{kV}$. The charging time was 2min. The distance between the needle and the mesh and between the mesh and the ground plate was 5cm and 2cm, respectively. The area of the mesh and the sample

was 78cm² and 25cm², respectively. The temperature and relative humidity is 293K and 30%RH.

After corona charging, the sample was quickly moved with the grounded electrode to the 6000B-14 high-resolution probe to measure the surface potential, the distance between the probe and sample is 5mm. The results were acquired with Trek 347B non-contacting voltmeters. The data were obtained through the LabView acquisition system.

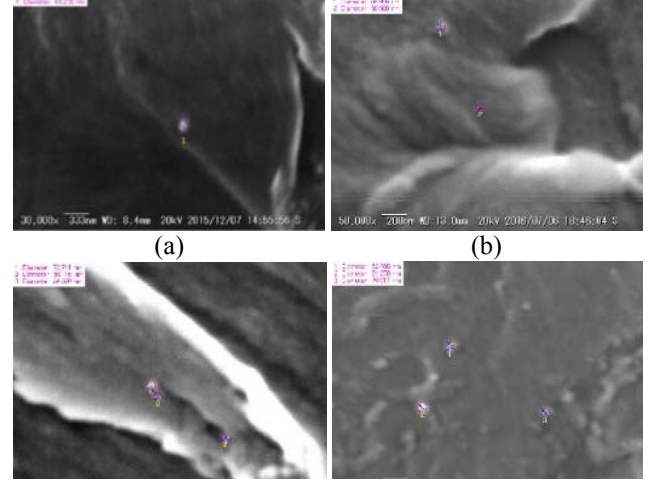


Fig. 1. SEM diagram of PI composites. (a-d) shows the 1wt%, 2wt%, 3wt%, 5wt% nano-ZnO modified PI specimen respectively

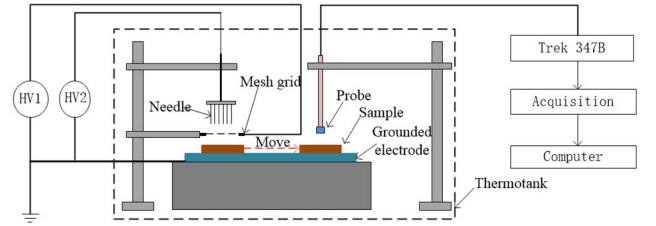


Fig. 2. Schematic diagram of the experimental setup for SPD

IV. RESULTS AND DISCUSSION

SPD under the trap energy distribution reveals the charging and discharging properties of the dielectric. The surface potential decay of negative or positive for PI is measured. Then, based on the potential decay measuring method and related theory, electron and hole trap energy distributions are obtained

A. SPD results of positive/negative corona discharge

The normalized SPD after negatively and positively corona charging results are shown in Fig. 3 and Fig. 4, it can be seen that before 5000s is the fast decay phase of the potential decay, and the decay of the potential is slowed after 5000s distinctly. Unexpectedly, pure PI has a higher rate of potential decay and eventually decays to 84% of the original potential. The decay rate and potential amplitude of the nano-ZnO modified PI in the rapid decay phase are significantly slower than that of the pure PI, the negative potential decay rate of the PI material increased with the increase of the ZnO mass fraction. The data

of 2wt% and 3wt% were close and there was a significant decrease in the 5wt%. The difference in the positive potential decay of 1wt% to 3 wt% is small, and there is almost no difference in the fast decay phase. In the slow decay phase, the decay of 2 wt% modified PI is minimal. The 5wt% decay rate and decay potential are greater than other mass fraction modified PI specimens but much smaller than that of pure PI.

During the potential decay process, the trapped charge may polarize the molecules in its vicinity and confined to the extremely unstable equilibrium state formed by the combined action of the externally applied electric field and the polarized charge. The external mechanical, thermal, and electric field disturbances may destroy this state and cause the charge to detrapp. After the withdrawal of the applied voltage, the above equilibrium state is broken. The release of polarized energy causes the surface charge to get energy out of the trap. The release of the charge causes the relaxation of dielectric and a large amount of charge to be detrapped. In the initial stage of dissipation, the homocharge density is high, and the heterocharge is easily attracted and recombined, and the speed of charge dissipation is fast. As the relaxation process tends to balance, the released polarization energy decreases, the amount of detrapped electron and hole decreases, the recombination weakens and the rate of charge dissipation slows down. For the reason that the mass fraction of ZnO affects the potential decay and the trap characteristics are shown in Section B.

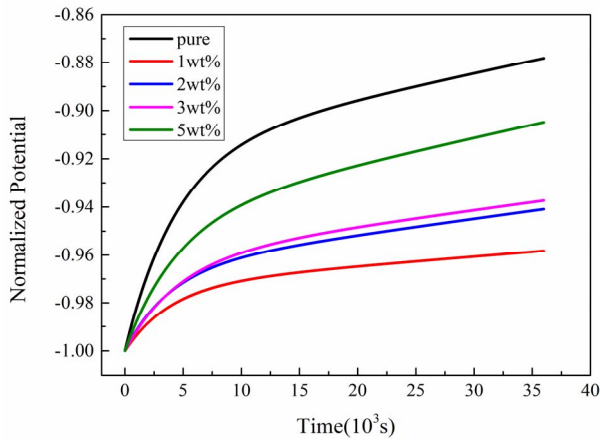


Fig. 3. Normalized surface potential decay after negatively charging

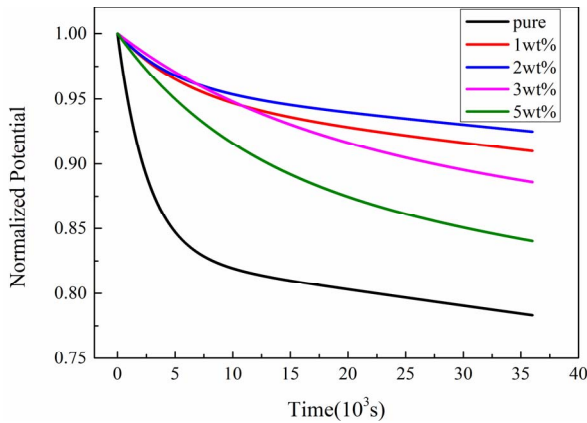


Fig. 4. Normalized surface potential decay after positively charging

B. Traps distribution characteristics

As shown in Fig. 5, the trap level of the electron trap changes rarely. As the mass fraction of ZnO increases, the trap charge density increases, but the trap density is less than that of pure PI. The electron trap energy level changes little in the region of 1.038-1.047eV. Fig. 6 shows that pure PI has the highest hole trap density and its energy level is $8.6 \times 10^{20} \text{ eV}^{-1} \cdot \text{m}^{-3}$ and 1.022eV. After the modification, the trap energy level rises to a certain extent, diverse between 1.047-1.08 eV. The trap density increases linearly with the mass fraction of ZnO. Combined with the TSC data in Fig. 7, the deep trap density of modified PI is lower than that of pure PI is confirmed. The calculated trap density is 10^{20} orders of magnitude, which is similar to the trap density estimated by the hopping conduction. In Fig. 7, the deep trap density calculated by TSDC method is the magnitude of 10^{18} , the shallow trap density is the magnitude of 10^{17} . The trap density shows different trends between shallow trap density and deep trap density. The deep trap density of SPD and TSDC results has the similar trend, that is the pure PI has the largest trap density. However, the shallow trap density of of pure PI is the lowest, the shallow trap density increases linearly with the mass fraction of ZnO.

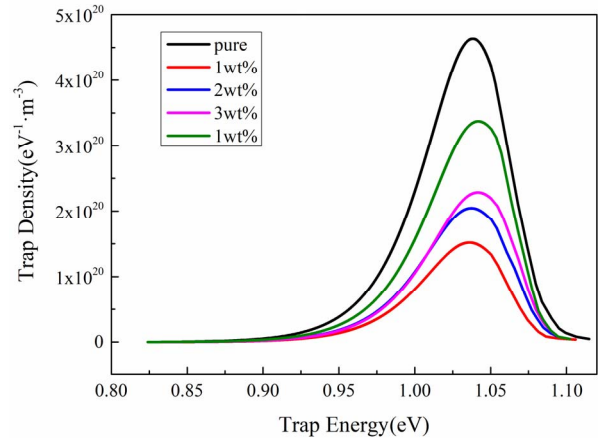


Fig. 5. Electron trap density distribution

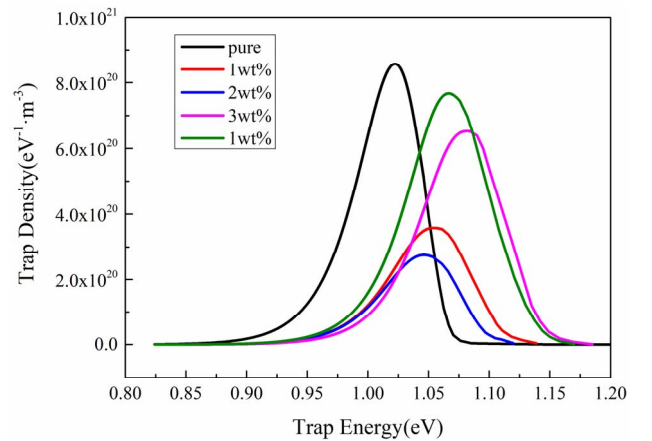


Fig. 6. Hole trap density distribution

It can be found that PI and its modified samples do not show the double-peak results of deep traps and shallow traps. Theoretical analysis and experimental results have shown that the space charge formed in the dielectric is almost concentrated on the surface of the sample at a level of 1 to 2 μm [10-12]. The surface potential of the dielectric is mainly determined by the surface layer. Therefore, the measurement result of the specimens in this paper mainly reflects the trap condition of the surface layer of the material, which is affected by the surface trap condition. Theoretically, the SPD method can obtain trap distribution between the conduction band and valence band. However, owing to the limitation of the experimental system and experimental methods, the decay time of holes and electrons at the shallow trap level of the material surface is too short to measure. Therefore, the initial data of the SPD cannot be obtained, namely, the trapped charge of shallow energy level cannot be reflected. However, the electronic surface states are usually deep traps in the forbidden energy band [13].

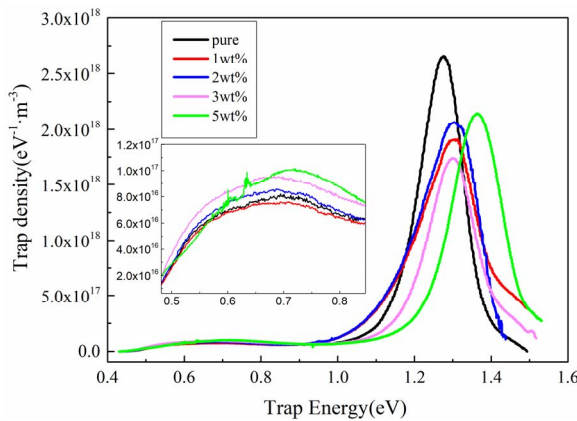


Fig. 7. TSC results of trap distribution

It is generally believed that the addition of nano-ZnO will bring in more interfaces and traps and increase trap density, but the test results in this paper did not quite fit. Tanaka concluded that the deeper traps can be formed in the second layer of the multi-core model for PI nanocomposites. The second layer is an interfacial region consisting of polymers chains strongly bound and/or interacted to the silane coupling agent which is bond to the nano-ZnO. The introduction of interface is the key to the improvement of trap depth. Nano-ZnO can reduce the density of deep traps in PI and increase the density of shallow traps through nucleation and interface action.

V. CONCLUSION

The paper referenced an SPD model considering the charge trapping and detrapping law of DC charging dielectric. The established SPD model can characterize the distribution of trap charge in PI well.

The ZnO modification results in a significant decrease in surface potential decay rate and decay amplitude, and the deep trap density also decreases to some extent. The deep electron trap has less change in energy level, while the deep hole trap energy level of ZnO modified PI is significantly higher than

that of pure PI. With the increase of the modified mass fraction, the surface potential decay rate and decay amplitude increase, and the deep trap density also increase to some extent. This method has a limitation of getting the shallow trap, but the deep trap of both hole trap and electron trap are reliable. By TSDC method, we found that the nano-ZnO introduced more shallow trap and decreased the deep trap.

ACKNOWLEDGMENT

The research work at the Xi'an Jiaotong University is supported by the National Nature Science Foundation of China – Surface Charging Mechanism of Non-linear Conduction Modified Polyimide under Simulated Low Earth Orbit, project No. 51607134

REFERENCES

- [1] Baker DN, "The occurrence of operational anomalies in spacecraft and their relationship to space weather," IEEE Trans Plasma Sci, vol. 28, pp.2007–2016, Dec 2000.
- [2] Simmons J G,Tam M C,"Theory of isothermal currents and the direct determination of trap parameters in semiconductors and insulators containing arbitrary trap distributions,"Phys. Rev. B,vol. 7, pp. 3706-3713, 1973.
- [3] Simmons J G,Taylor G W, "High-field isothermal currents and thermally stimulated currents in insulators having discrete trapping levels," Physical. Review. B., vol. 5, pp. 1619-1629, 1972.
- [4] Chenyang Liu, Xiaoquan.Zheng, Ping Peng, "The Nonlinear Conductivity Experiment and Mechanism Analysis of Modified Polyimide (PI) Composite Materials With Inorganic Filler," IEEE Transactions on Plasma Science, vol. 43, pp. 3727-3233, Oct 2015.
- [5] Zhen-Jun Zhang, Xiao-Quan Zheng, Jiang Wu, Wen-Bin Wu, and Pei Yang, "DC surface flashover characteristics of polyimide in vacuum under electron beam irradiation," IEEE Trans. Dielect. Electr. Insul, vol. 22, pp. 604-610, Feb 2015.
- [6] D. Das-Gupta, "Surface charge decay on insulating films," in Int'l. Sympos. Electr. Insul., vol. 2, pp. 296-299, 1988.
- [7] P. Molinie and M. Goldman, "Evolution of the surface layer appearing with the corona ageing of epoxy samples," IEEE. Int'l. Conf. Solid Dielectrics (ICSD), pp. 482-486, 1995.
- [8] P. Lloveraz and P. Molinie, "New methodology for surface potential decay measurements: application to study charge injection dynamics on polypropylene films," IEEE Trans. Dielect. Electr. Insul., vol. 11, pp. 1049-1056, Dec 2004.
- [9] Wen-Wei Shen, Hai-Bao Mu, Guan-Jun Zhang, Jun-Bo Deng, and De-Min TuHai-Bao Mu, "Identification of electron and hole trap based on isothermal surface potential decay model," Journal of Applied Physics,vol. 56, pp. 083706, 2013.
- [10] Kurtz S R,Anderson R A, "Properties of the metal-polymer interface observed with space-charge mapping techniques," Journal of Applied Physics,vol. 60, pp. 681-687, 1986.
- [11] Torpey P A, "Double-carrier injection and recombination in insulators including diffusion effects," Journal of Applied Physics,vol. 56, pp. 2284-2294, 1984.
- [12] Zhang G J, Wang X R,Yan Z, "Analysis on surface trapping parameters of solid dielectrics," 7th International Conference on Solid Dielectric (ICSD),Eindhoven,pp. 236-239, 2001.
- [13] Molinie, P, "A review of Mechanisms and models accounting for surface potential decay," IEEE Transactions on Plasma Science, vol. 40, pp. 167-176, Feb 2012.